

# Danesh Morales Hashemi

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Updated on May 15, 2026

## Research Positions

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- 2025 – Present**    **Graduate Research Assistant**, Institute for Quantum Computing  
Advisor: [Graeme Smith](#)
- 2024 – 2025**    **Undergraduate Research Assistant**, Institute for Quantum Computing  
Advisor: [Joseph Emerson](#)  
Topic: *Circuit Benchmarking for Efficient and Scalable Fidelity Estimation*
- Fall 2023**    **Undergraduate Research Assistant**, University of Waterloo  
Advisor: [Roberto Guglielmi](#)  
Topic: *Controllability of Fokker-Planck Equation*

## Education

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- 2025 – Present**    **MMath in Applied Mathematics**, University of Waterloo  
Quantum Information Specialization
- 2020 – 2025**    **BMath in Mathematical Physics**, University of Waterloo  
Co-op Program · Pure Mathematics Minor · Graduated with [Dean's Honours](#)

## Conferences and Talks

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- March 2025**    **“Demonstrating Circuit Benchmarking for Scalable and Efficient Fidelity Estimation”**  
**APS March Meeting 2025**, Session on Scalable Verification and Validation (QCVV), Anaheim Convention Center

## Teaching Experience

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- Spring 2025**    **Teaching Assistant**, AMATH 250 – Introduction to Differential Equations
- Fall 2023**    **Instructional Support Assistant**, MATH 115 – Linear Algebra for Engineering
- Winter 2022**    **Instructional Support Assistant**, CS 115 – Introduction to Computer Science

## Projects

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- December 2025**    **“Introduction to Cycle Benchmarking for Quantum Error Diagnostics”**  
QIC 820 Final Project, Institute for Quantum Computing
- December 2025**    **“Predicting MLB Playoffs Using Quantum Machine Learning”**  
QIC 890 Final Project, Institute for Quantum Computing.  [Link](#)

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|--------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>April 2025</b>  | <b>“Demonstrating Circuit Benchmarking for Efficient and Scalable Fidelity Estimation”</b><br>Undergraduate Thesis, University of Waterloo                 |
| <b>August 2024</b> | <b>“Introduction to Superoperator Representations and Characterization of Completely Positive Maps”</b><br>AMATH 900 Final Project, University of Waterloo |

## Awards and Scholarships

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|-----------------------------------------------------------------------------|--------------------|
| Institute for Quantum Computing Undergraduate Research Award 2024           | <b>16,000 CAD</b>  |
| University of Waterloo President’s Scholarship 2020                         | <b>2,500 CAD</b>   |
| Full Undergraduate Educational Excellence Scholarship 2019, SENACYT, Panama | <b>250,000 USD</b> |

## Relevant Coursework (with links to syllabus)

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|--------------------------------|-------------------------------|-------------------------|
| Quantum Physics                | Quantum Theory 1/2/3          | Representation Theory   |
| Quantum Information Processing | Quantum Machine Learning      | Groups and Ring Theory  |
| Quantum Devices                | Theory of Quantum Information | Galois and Field Theory |